

concise batches. Traveling is a nuisance fraught with all manner of physical danger; you will settle within a narrow radius of where you were born unless famine or some invading uncivilized tribe dislodges you and your relatives from your happy settlement. The number of people you would get to know in a lifetime will be small. Should a crime be committed, it will be easy to gauge the evidence of guilt within the small number of possible suspects. If you are unjustly convicted of a crime, you will argue in simple terms, propounding simple evidence like "I was not there as I was praying in the temple of Baal and was seen at dusk by the high priest" and add that Obedshemesh son of Sahar was more likely to be guilty because he had more to gain from the crime. Your life would be simple, hence your space of *probabilities* would be narrow.

The real problem is, as I have mentioned, that such a natural habitat does not include much information. An efficient computation of the odds was never necessary until very recently. This explains why we had to wait until the emergence of the gambling literature to see the growth of the mathematics of probability. Popular belief holds that the religious backdrop of the first and second millennium blocked the growth of tools that hint at absence of determinism, and caused the delays in probability research. The idea is extremely dubious; we simply did not compute probabilities because we did not *dare* to? Surely the reason is rather because we did not *need* to. Much of our problem comes from the fact that we have evolved out of such a habitat faster, much faster than our genes. Even worse; our genes have not changed at all.

KAFKA IN A COURTROOM

The O.J. Simpson trial provides an example of how our modern society is ruled by probability (because of the explosion in information), while important decisions are made without the smallest regard for its basic laws. We are capable of sending a spacecraft to Mars, but we are incapable of having criminal trials managed by the basic laws of probability – yet evidence is clearly a probabilistic notion. I remember buying a book on probability at a Borders Books chain bookstore only a short distance from the Los Angeles courthouse where the "trial of the century" was taking place – another book that crystallized the highly

sophisticated quantitative knowledge in the field. How could such a leap in knowledge elude lawyers and jurors only a few miles away?

People who are as close to being criminal as probability laws can allow us to infer (that is with a confidence that exceeds the *shadow of a doubt*) are walking free because of our misunderstanding of basic concepts of the odds. You could be convicted for a crime you never committed owing to a poor reading of probability – for we still cannot have a court of law properly compute the joint probability of events (the probability of two events taking place at the same time). I was in a dealing room with a TV set turned on when I saw one of the lawyers arguing that there were at least four people in Los Angeles capable of carrying O.J. Simpson's DNA characteristics (thus ignoring the joint set of events – we will see how in the next paragraph). I then switched off the television set in disgust, causing an uproar among the traders. I was under the impression until then that sophistry had been eliminated from legal cases thanks to the high standards of republican Rome. Worse, one Harvard lawyer used the specious argument that only 10% of men who brutalize their wives go on to murder them, which is a probability unconditional on the murder (whether the statement was made out of a warped notion of advocacy, pure malice, or ignorance is immaterial). Isn't the law devoted to the truth? The correct way to look at it is to determine the percentage of murder cases where women were killed by their husband *and* had previously been battered by him (that is, 50%) – for we are dealing with what is called *conditional* probabilities; the probability that O.J. killed his wife *conditional* on the information of her having been killed, rather than the *unconditional* probability of O.J. killing his wife. How can we expect the untrained person to understand randomness when a Harvard professor who deals and teaches the concept of probabilistic evidence can make such an incorrect statement?

More particularly, where jurors (and lawyers) tend to make mistakes, along with the rest of us, is in the notion of joint probability. They do not realize that evidence compounds. The probability of my being diagnosed with respiratory tract cancer and being run over by a pink Cadillac in the same year, assuming each one of them is 1/100,000, becomes 1/10,000,000,000 – by multiplying the two (obviously independent) events. Arguing that O.J. Simpson had 1/500,000 chance of not being the killer from the blood standpoint (the lawyers used the